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Title of chapter: Chapter 06 - Detection theory

Summary:

Chapter 06 covers detection theory, which serves as the foundation for evaluating the basic performance of machine learning models in binary decision problems like loan approvals. It highlights that models are decision functions which map input features to predictions, and that their performance must be measured using metrics that reflect the real-world costs and risks of decisions. The chapter describes how to create a confusion matrix and related probabilities (true positives, false positives, true negatives, and false negatives), as well as how to convert these into metrics such as accuracy, precision, recall, F1-score, and the ROC curve. It also introduces the notion of Bayes risk, which takes into account the costs of different error types and gives a theoretically optimal performance standard or benchmark for any decision function.

Confusion Matrix:

	$Y = 1$	$Y = 0$
$\hat{y}(X) = 1$	TP	FP
$\hat{y}(X) = 0$	FN	TN

$P(\hat{y}(X) Y)$	$Y = 1$	$Y = 0$
$\hat{y}(X) = 1$	p_{TP}	p_{FP}
$\hat{y}(X) = 0$	p_{FN}	p_{TN}

$P(Y \hat{y}(X))$	$Y = 1$	$Y = 0$
$\hat{y}(X) = 1$	p_{PPV}	p_{FDR}
$\hat{y}(X) = 0$	p_{FOR}	p_{NPV}

As you can gather from the tables,

$$p_{TP} \approx n_{TP} / (n_{TP} + n_{FN})$$

$$p_{FN} \approx n_{FN} / (n_{FN} + n_{TP})$$

$$p_{FP} \approx n_{FP} / (n_{FP} + n_{TN})$$

$$p_{TN} \approx n_{TN} / (n_{TN} + n_{FP})$$

$$p_{PPV} \approx n_{TP} / (n_{TP} + n_{FP})$$

$$p_{FOR} \approx n_{FN} / (n_{FN} + n_{TN})$$

$$p_{FDR} \approx n_{FP} / (n_{FP} + n_{TP})$$

$$p_{NPV} \approx n_{TN} / (n_{TN} + n_{FN})$$

$$F1 = 2 * p_{TP} * p_{PPV} / (p_{TP} + p_{PPV})$$

ROC Curves and AUC: Plot true positive rate vs false positive rate across thresholds; AUC is a threshold-independent measure of performance.

PR Curves (Precision-Recall): Useful when positive class cases are uncommon.

This chapter emphasizes that metric selection must be coordinated with problem goals (for example, in loans, the costs of false approvals versus false denials differ).

Considering costs, the description of performance for the decision function is known as the Bayes risk (R):

$$R = (c_{10} - c_{00}) * p_0 * p_{FP} + (c_{01} - c_{11}) * p_1 * p_{FN} + c_{00} * p_0 + c_{11} * p_1$$
$$\text{Therefore, } R = c_{10} * p_0 * p_{FP} + c_{01} * p_1 * p_{FN}$$

Formally, the Bayes-optimal decision function is:

$$\hat{y}^*(\cdot) = \operatorname{argmin}_{\hat{y}} E[c(Y, \hat{y}(X))]$$

When simplified, it takes the form of:

$$\hat{y}^*(\cdot) = 0, \text{ when } \Lambda(x) \leq \eta \text{ or } 1, \text{ when } \Lambda(x) > \eta,$$
$$\text{where the likelihood ratio } \Lambda(x) = p_{X|Y}(x | Y = 1) / p_{X|Y}(x | Y = 0)$$
$$\text{and the threshold } \eta = c_{10} * p_0 / c_{01} * p_1$$

The chapter further distinguishes binary classification from probabilistic risk assessment, emphasizing the importance of calibration when the outputs are continuous scores representing the likelihood of an occurrence. Properly calibrated models enable decision-makers to interpret scores as meaningful probabilities, which is essential in high-risk domains. In risk assessment, interpretability and trust are particularly important since the model's output influences human decisions. Even if a model's AUC is high, it might mislead decision-makers. As a result, evaluation must extend beyond accuracy or ROC/AUC to incorporate calibration plots, Brier scores, and expected cost analysis.

The Brier score is an appropriate performance metric for the continuous-valued output score of the decision function:

$$\text{Brier score} = E[(S - Y)^2]$$

Detection theory, paired with Bayes-optimal analysis, assists data scientists in understanding the limitations of model performance, selecting relevant metrics in line with business objectives, and evaluating models across several operational points to make informed, risk-sensitive decisions.

What are two points that you learned after reading this chapter?:

- Bayes risk determines the best possible performance for the given problem and data distribution theoretically, and it serves as a benchmark against which to assess any model.

- Calibration is important when working with probabilistic outputs since it guarantees that predicted probabilities match the observed outcomes and allows for risk-informed decision-making.

What are two questions that you have about this chapter?:

- How can one realistically estimate the Bayes-optimal risk in complicated, high-dimensional real-world scenarios where feature distributions are unknown?
- How can calibration be maintained or improved when deploying models in dynamic situations where the underlying data distribution changes over time?

Do you have some initial thoughts or insights to approach the questions?:

- One technique to determining Bayes-optimal risk may be to utilize simulations or approximation models based on existing data to estimate likelihoods and priors, while keeping in mind that this only provides an upper bound rather than a definite limit. Another approach is semi-parametric modeling, which makes assumptions about distributions to simplify computations.
- To maintain calibration over time, we could use continuous monitoring of predicted probabilities vs. observed outcomes and apply online recalibration methods (e.g., isotonic regression or Platt scaling) when drift is detected. Retraining models with new data on a regular basis can also help maintain reliable calibration.